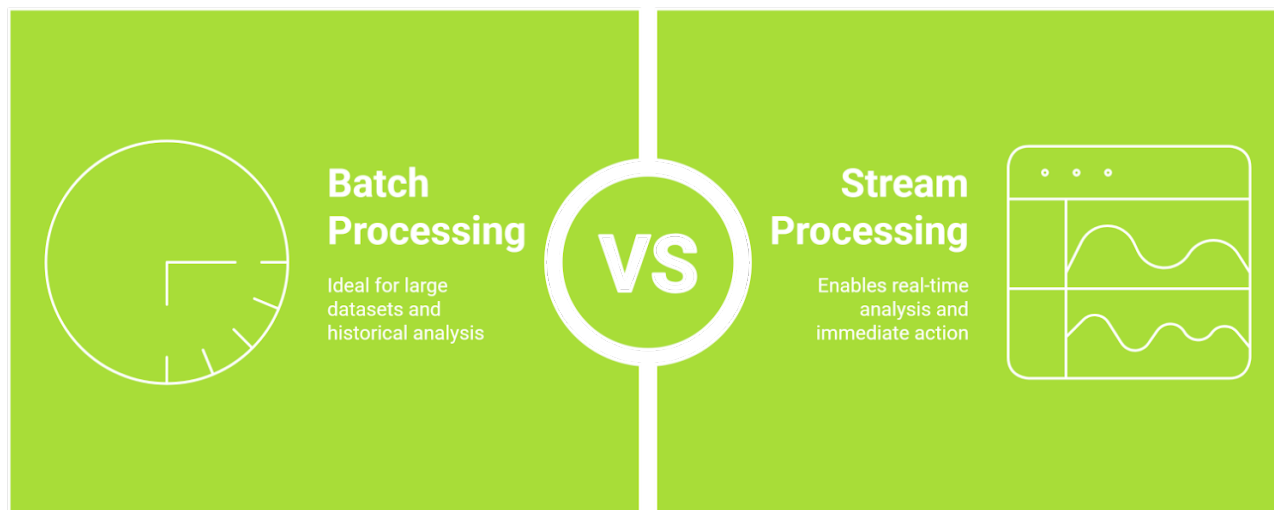


Batch Processing or Streaming: What's Better?

Are you debating whether to go for batch processing your company's data or streaming it in real time? Here's a look at the trade-offs involved when selecting which process is best for your architecture and business, with hybrid models emerging as winners.



Data flows in many directions daily — financial transactions, visitors clicking on websites, sensor data feeds, mobile app activity, and more. When businesses adopt data-driven practices, one of the first decisions they face is whether to process that data in real time or as a batch.

Batch processing is the old way of handling data; it is used when you have a lot of data and need to process it at certain intervals. It is a great tool for end-of-day reporting, backups, or historical analytics. For decades, tools like Apache Hadoop, Apache Nifi, and more recently, Airflow, have been used to handle batch processes.

Stream processing, on the other hand, enables you to do something right now! It allows you to process information in real time, and thus create dashboards, alerts, and fast decisions. Open source stream processing tools

like Apache Kafka, Flink, and Spark Streaming are making the stream-first world easier.

The hard part is choosing what method and which open source tools are best for your specific data processing needs. There is a right answer, but first, we must understand when and why timeliness matters in data pipelines, how each approach has developed over time, and the realities of real time versus scheduled data workflows.

The importance of timing in data pipelines

Imagine you are operating an e-commerce site. A customer puts a product in their cart but abandons it. If your system can alert the customer within minutes with a discount, you may win that sale back. If you wait a day to process that data, the window for the sale is probably gone. That is the power of timing in data processing.

Data pipelines—whether they are dealing with customer behaviour, financial transactions, IoT sensor data, or emerging social media trends—are evaluated on the speed at which they can convert data to decisions. A pipeline that processes today's data would be too slow for fraud detection or tracking inventory one day behind, but fine for the next week's sales report.

Timing influences:

- Business agility
- Customer experience
- Operational effectiveness
- Infrastructure costs

From ETL to real time: The shift to streaming data integration

Traditionally, organisations used ETL (extract-transform-load). Data was extracted from a series of sources, cleaned up, fixed a little, structured, and loaded into warehouses at one time every night. This was batch processing.